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Duration	6 mins 5 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(log n)
b. O👎 ✓
c. O(n log n)
d. O(1)

Question **2**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. At the beginning of the list
b. In the left sublist
c. In the middle of the list
d. In the right sublist ✓

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There must be a mechanism to delete and/or insert elements in the list ✓
b. The list must be sorted
c. Number values should only be present
Number values should only be present
d. There should be direct access to the middle element in any sublist

Question **4**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 89 and 94

b. 94 and 99

c. 90 and 100

d. 90 and 99 ✓

Question **5**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. 1

b. n ✓

c. log n

d. n/2

Question **6**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is large and sorted

b. When the list is small and unsorted ✓

c. When the list is large and unsorted

d. When the list is small and sorted

Question **7**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(log n) ✓

b. O(1)

c. O(n log n)

d. O🥐

Question **8**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The last element is returned
b. An error is raised
c. The search is considered unsuccessful ✓
d. The first element is returned

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search divides the list into halves.
b. Linear search is more efficient than binary search on large lists.
c. Linear search can be applied to both sorted and unsorted lists. ✓
d. Linear search requires the list to be sorted.

Question **10**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By sorting the list first
b. By comparing each element in the list sequentially ✓
c. By using a hash function
d. By dividing the list into halves

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It does not work with negative numbers
b. It requires the list to be sorted ✓
c. It is slower than linear search for small lists
d. It can only be applied to large lists

Question **12**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search works on unsorted lists
b. Binary search has a lower time complexity on large, sorted lists ✓
c. Binary search can find multiple instances of the target element
d. Binary search does not require dividing the list

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Dividing the list in half repeatedly
b. Skipping every second element
c. Sorting the list before searching
d. Checking each element sequentially ✓

Question **14**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Hash search
b. Binary search
c. Interpolation search
d. Linear search ✓

Question **15**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By dividing the list length by two ✓
b. By comparing each element sequentially
c. By starting from the first element
d. By using a hash function